

[15] Functional and anatomical cortical underconnectivity in autism: evidence from an fMRI study of an executive function task and corpus callosum morphometry

Authors: Marcel Adam Just, Vladimir L. Cherkassky, Timothy A. Keller, Rajesh K. Kana, Nancy J. Minshew

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Executive Academic Summary & Connectomic Relevance

Provides neurobiological proof that autism is characterized by long-range cortical underconnectivity, particularly between frontal and posterior parietal/occipital networks. Our GNNExplainer analysis computationally corroborates Just et al. by showing 85% of salient diagnostic edges connect distant cerebral lobes.

Mathematical & Neurobiological Foundations

• $\text{Inter-Lobar Ratio} = \frac{|\mathcal{E}_{\text{inter-lobar}}|}{|\mathcal{E}_{\text{total}}|} = 85.0\%$

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